CYBERSECURITY PROJECT REPORT

PASSWORD STRENGTH ANALYZER

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A beginner-friendly Python project that checks basic password characteristics and provides suggestions for improving password security.

1. Abstract

Password Strength Analyzer is a beginner-level cybersecurity project developed using Python. It checks password length, uppercase and lowercase letters, digits, and special characters. It displays a basic strength rating and suggestions for improvement. The program processes the test password locally and does not save or transmit it.

2. Introduction

Passwords protect email, social media, educational, and workplace accounts. Short, reused, or easily guessed passwords can increase the risk of unauthorized access. This project demonstrates password-awareness principles through a simple command-line application.

3. Problem Statement

Users may choose passwords that are too short or lack variety. A simple tool can provide immediate feedback and encourage users to create longer, less predictable passwords.

4. Objectives

• Build a simple Python cybersecurity-awareness program.
• Check basic password characteristics.
• Provide a strength label and practical suggestions.
• Process the password locally without storing it.
• Encourage unique passwords and multi-factor authentication.

5. Tools and Technologies

• Python 3.8 or newer
• re and getpass modules from Python's standard library
• IDLE, VS Code, or another Python editor
• Optional: Jupyter Notebook

6. Algorithm

1. Start the program.
2. Ask the user to enter a made-up test password.
3. Check length, uppercase, lowercase, digits, and special characters.
4. Check a small list of commonly guessed passwords.
5. Calculate a score and classify it as Weak, Moderate, or Strong.
6. Display the rating and suggestions.
7. End the program.

7. Flowchart

START
↓
Enter a made-up test password
↓
Check length and character types
↓
Calculate score and strength
↓
Display rating and suggestions
↓
END

8. Implementation

The program uses Python regular expressions to check character types. Each passed check adds to the score. The score maps to a basic label: Weak (0–2), Moderate (3–4), or Strong (5–6). A short list of common passwords is included for demonstration.

9. Testing and Expected Output

Use made-up test values only. A short value should trigger a length suggestion; letters-only input should prompt for digits and special characters; a complex example such as Example!Pass123 should pass the basic checks; and a listed common password should trigger a warning.

Test case	Expected behaviour	Purpose
Short test value	Length suggestion	Check length rule
Letters only	Suggest digits/special characters	Check character rules
Example!Pass123	Strong rating under basic rules	Positive demonstration
password123	Common-password warning	Check common-password rule

10. Security and Privacy

The program does not write passwords to a file or send data over a network. Terminal input uses getpass so typed input is hidden in a typical terminal. Never enter a real account password into a classroom demonstration tool.

11. Limitations

This checker uses simple rules, not a comprehensive password-strength estimation method. Passing the checks does not guarantee security. The common-password list is small and the project does not query breached-password databases.

12. Future Enhancements

- Add a larger offline common-password list.
- Add a graphical user interface.
- Include unit tests.
- Add a more advanced strength-estimation method.

13. Conclusion

The Password Strength Analyzer demonstrates an introductory cybersecurity concept using simple Python code. It gives feedback on basic password characteristics and promotes better security habits. It is an educational aid, not a replacement for a password manager, multi-factor authentication, or organizational security controls.

14. References

Python getpass documentation: https://docs.python.org/3/library/getpass.html
Python regular expressions documentation: https://docs.python.org/3/library/re.html
OWASP Password Storage Cheat Sheet:
https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html